

Graded Containment with an Observability Price: Sequential Response to Ambiguous Trading Incidents

Chiheb Mejrissi

September 29, 2026

Abstract

Draft pending. Claim: a belief-based graded policy dominates a threshold kill-switch, and containment has an observability price: below a critical retained discriminability, throttling is counterproductive.

1 Introduction

Planned. Containment changes future observations; Paper 1’s Theorem 2 predicts reduced discriminability when throttling suppresses state feedback.

2 Related work

Planned. POMDP intrusion response, optimal stopping, kill-switch and circuit-breaker practice.

3 Model

Planned. Belief-grid POMDP, actions monitor/throttle/verify/kill; two-state baseline and three-state (mixed-cause) extension. Code: `src/paper3_pomdp`.

4 Structure of the optimal policy

Planned. Policy maps; dominance over the kill-switch; critical κ^* .

5 Numerical study

Planned. Equivalent-Gaussian d from Paper 1 as a calibration heuristic, plus a direct sweep over d .

6 Discussion

Planned.

7 Conclusion

Planned.

References

- [1] Kim Hammar and Rolf Stadler. Intrusion prevention through optimal stopping. *IEEE Transactions on Network and Service Management*, 19(3):2333–2348, 2022.
- [2] Erik Miehl, Mohammad Rasouli, and Demosthenis Teneketzis. A POMDP approach to the dynamic defense of large-scale cyber networks. *IEEE Transactions on Information Forensics and Security*, 13(10):2490–2505, 2018.